Two experimental studies of children's comprehension of presuppositional *you* 'again' in Mandarin

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Abstract

Previous studies of children's comprehension of the presupposition of English "again" and its Mandarin equivalent "you" (requiring that an event of the same kind previously took place) have reported successful acquisition by preschool age. Such studies have generally used tasks that require children to make binary judgments (e.g., 'true'/'false', 'yes'/'no'). However, sentences involving presupposition failure are not necessarily judged as 'false' by adults, making such tasks less than optimal for assessing children's knowledge of presuppositions. In this study, we examine Mandarin-acquiring children's comprehension of the presupposition of "you" ('again'), using two alternative tasks that target the presupposition without requiring participants to make binary judgments about sentences involving presupposition failure. In the Question-Answer Task (Experiment 1), participants listened to stories and answered a *wh*-question after each story. The results revealed that most children responded as though they ignored the presupposition trigger "you", which highlights the methodological challenge of investigating children's knowledge of presupposition. We then conducted a Felicity Judgment Task (Experiment 2), which highlighted the presence of the trigger by asking participants to decide which of two speakers—one who produced "you" and one who did not—answered better. In this task, children displayed sensitivity to the presupposition of "you", albeit not at adult-like levels. We discuss the implications of these findings in relation to children's knowledge of "you" and its obligatory presence when its presupposition is satisfied, a potential ambiguity in its interpretation, and possible differences between children and adults in how they project presuppositions from *wh*-questions.

Keywords: presupposition, again, preschoolers, Question-Answer Task, Felicity Judgment Task

1. Introduction

This study concerns children's acquisition of presupposition, a kind of backgrounded information that a speaker takes for granted when making an utterance (Stalnaker, 1973), in contrast to information that is explicitly asserted. The question of how children recognize and process this type of linguistic inference has received increasing attention in the field of language acquisition. Our focus is on the repetition-denoting adverb "again", which triggers the presupposition that a prior event of the same kind has previously occurred, as illustrated by the English example (1) and its Mandarin translational equivalent (2).

(1) Mary sneezed again.

(2) Mali **you** da-le penti.[1]
Mary again hit-ASP sneeze
'Mary sneezed again.'

Presupposition of (1) and (2): Mary sneezed at an earlier time.

We can test the status of the presupposition by using standard presupposition projection tests (see Langendoen & Savin, 1971). For example, the presupposition that Mary sneezed before is preserved in polar questions, as shown in the English example (3) and its Mandarin counterpart (4) (where *ma* is a sentence-final particle used in polar questions).

(3) Did Mary sneeze again?

(4) Mali **you** da-le penti ma?
Mary again hit-ASP sneeze SFP
'Did Mary sneeze again?'

Presupposition of (3) and (4): Mary sneezed at an earlier time.

"Again" has accordingly been characterized as a presupposition trigger. According to some analyses (e.g., Kamp & Rossdeutscher, 1994; von Stechow, 1996), the adverb is truth-conditionally vacuous, contributing only to the presupposition of the sentence in which it appears. In other words, a sentence with "again" has the same asserted content as the equivalent sentence without it, but additionally presupposes a prior occurrence of the modified event. Under such an analysis, (1) and (2) assert that Mary sneezed, and presuppose that Mary sneezed at an earlier time.

In the current study, we employed two tasks to investigate Mandarin-speaking children's comprehension of "you" ('again') (2), one of the translational equivalents of English "again".[2,3] The first experiment employed a Question-Answer Task, in which participants listened to short stories and answered a *wh*-question after each story. To anticipate, while this task avoids the problem of asking participants to make binary judgments about sentences with presupposition failure, the results run contrary to those of previous studies: preschool-aged children do not appear to interpret Mandarin "you" in an adult-like manner. To address some potential explanations for this finding, we next conducted a Felicity Judgment Task, which highlighted the presence of "you" by asking participants to decide which of two speakers—one who produced "you" and one who did not—provided a better answer. To preview, the results from this task indicate that preschool-aged children successfully understand the presuppositional trigger (albeit not at fully adult-like levels). Overall, the findings from the two experiments suggest that preschool-aged children possess nuanced knowledge of the presupposition of "you" ('again'), and moreover that the non-

[1] Throughout the paper, we use boldface to indicate stress.
[2] Apart from "you", Mandarin "zai" can also denote repetition, though it has a different distribution (see Lin & Liu, 2009; Liu & Yip, 2023; Lü, 1999 for discussion). We restrict the scope of our study to "you", leaving "zai" for future investigation.
[3] In addition to denoting repetition, Mandarin "you" has other, less frequent interpretations, such as temporal continuation, addition, and rhetorical uses (see Lü, 1999). There is some controversy regarding the core meaning of "you" and how many distinct readings of "you" there are (see Biq, 1988; Lü, 1999; Shao & Rao, 1985; Shi, 1990), but the repetition-denoting meaning is the most typical and salient interpretation, particularly when "you" is stressed. We thus set aside these other readings of "you" in the current paper, as they fall outside the scope of the study.

adultlike behavior observed in the Question-Answer Task should not be attributed to a lack of knowledge.

Before moving on to our study, we will first motivate our choice to focus on children's acquisition of the trigger "again", based on methodological, theoretical, and practical considerations. First, previous comprehension studies of children's acquisition of this presupposition trigger report successful acquisition by preschool age (for English, see Xu & Snyder, 2017; for Mandarin, see Liu, 2009, 2015; Liu et al., 2011; Xu et al., 2022, 2024). However, upon closer examination of some of these studies, there may be methodological reasons to question some of the findings. Specifically, most prior comprehension studies on this trigger have employed tasks that require participants to make binary judgments (e.g., answering a yes-no question containing "again" or judging the truth or falsity of a declarative sentence with the trigger when its presupposition is not satisfied), as will be described in further detail in the next section. However, the presuppositional nature of "again" presents a methodological challenge for such tasks, since presupposition failure does not necessarily elicit in adults a clear intuition that the sentence is 'false', but may instead evoke a "squeamish" feeling that the sentence is neither true nor false (Strawson, 1950, 1964; von Fintel, 2004; among others). For example, in a context in which Mary did not previously sneeze, it is intuitively difficult to assess what the truth value of the sentences (1) and (2) should be. Some theories characterize this as a truth value gap (e.g., Heim & Kratzer, 1998; Strawson, 1950, 1964; von Fintel, 2004). In tasks that elicit binary judgments from adults, some participants might group presupposition failures with a 'false' response; others might find it impossible to assign a truth value, judging such sentences as neither completely true nor completely false (see Zehr, 2014 and Križ & Chemla, 2015 for experimental evidence of this latter pattern for presupposition failures and undefined cases of homogeneity, respectively). Because it is unclear how adult participants should respond in such truth value gap scenarios, the binary judgment tasks employed in previous studies may be less suitable for assessing children's knowledge of presupposition.

Second, given that the semantic contribution of "again" is merely presuppositional (as opposed to triggers like "only" which contribute both presuppositional and assertive meanings), it is ideal for examining how children attend to non-at-issue information in the discourse. "Again" is also intriguing from a developmental perspective in that it conveys a meaning that is similar to that of expressions like "more than once"/"twice", which carry no presupposition. This gives rise to the question of how learners determine what meanings are presupposed as opposed to asserted.

Finally, from a practical perspective, the target presupposition of "you" (namely that an event of the same kind occurred previously) is one that is relatively easy to implement in visually illustrated stories of the kind children are used to seeing; it thus provides a feasible test case for understanding the development of presuppositions more broadly.

The rest of the paper is organized as follows. Section 2 provides a brief introduction to the relevant background. Sections 3 and 4 present the two experiments. Section 5 discusses the implications of the findings, as well as directions for future research.

2. Background

2.1. Previous studies on the acquisition of "again"

Developmental studies on the acquisition of "again" are relatively limited. Spontaneous production data reveal that children begin using "again" felicitously to denote repetition before the age of 3;06 (Lee, 2005; Penner et al., 2000; Xu & Snyder, 2017; Xu et al., 2024). However, these results are based on rather small sample sizes: three Cantonese-acquiring children in the study by Lee (2005), four English-acquiring children in the study by Xu and Snyder (2017), and two Mandarin-acquiring children in the study by Xu et al. (2024).[4]

Liu (2009, 2015) and Liu et al. (2011) present a series of experiments investigating children's knowledge of repetition-denoting "you", using both production and comprehension measures. In their elicited production task, 17 older children (age range: 2;06-3;10) and 5 younger children (age range: 2;00-2;06) were presented with contexts in which an event (e.g., a boy tossing a certain ball; something falling onto a table) occurred twice.[5] Each time an event took place, they were asked what happened. The authors report that 2- and 3-year-old children were able to correctly produce "you" when they saw an event happen for the second time.

To assess children's comprehension of "you", the researchers used different tasks for participants of different age groups. For younger children (aged 2;00-3;06, n=14, with 7 two-year-olds and 7 three-year-olds), the researchers designed an interactive, play-based task. In each trial, the child would first perform an action, e.g., putting a toy car in a box. The investigator would then ask the child, "What did you put in the box?" After the child answered, the investigator would ask, "Ni shi-bu-shi you fang-le chezi jinqu a?" ('Did you put a car in the box again?') (using stressed "you"). A response of 'no' or carrying out the action a second time (e.g., putting another car in the box) was taken as evidence that the child understood the meaning of stressed "you". The authors report correct response rates of 74.3% for three-year-olds and 60% for two-year-olds, with most children either answering 'no' or repeating the action (though the authors do not report the proportion of each specific response type). Based on these results, the authors conclude that children possess knowledge of repetition-denoting "you" from an early age. However, there may be reasons to be cautious in drawing this conclusion. First, if the presupposition is not satisfied (i.e., the child had not already previously put a car in the box), it is not obvious whether the answer to the investigator's polar question should be 'yes' or 'no' (given the forced binary choice). This makes the polar question unnatural in the experimental set up, which could explain why some child participants gave no response. Second, children who interpreted the question as a request might have repeated the action because they understood the question as an instruction to 'put a car in the box' rather than genuinely understanding the presupposition of "you".

For children aged 3;06 to 3;11 (n=9), the researchers used a Truth Value Judgment Task (TVJT) (Crain & Thornton, 1998), with adult participants serving as controls. Participants listened to short stories in which an event occurred only once (e.g., a panda ate a banana), and then were asked to judge a puppet's statement containing "you", e.g., "Xiongmao you chi-le xiangjiao" ('The panda ate a banana again'). Like adults, 7 out of 9 child participants rejected such sentences (with an overall rejection rate of 80%), which led the researchers to conclude that the children understood the repetition-based meaning of "you". Note, however, that this conclusion is based on a rather small sample size.

Xu (2015), Xu and Snyder (2017), and Xu et al. (2022, 2024) also used the TVJT to examine English- and Mandarin-acquiring children's knowledge of "again"/"you". These studies were focused on the so-called restitutive interpretation of "again" (e.g., Beck, 2005; Beck & Johnson, 2004; von Stechow, 1995, 1996), which is not relevant for our study, but they did include a control condition that tested the repetition-denoting meaning. In their TVJTs, participants listened to a

[4] Penner et al. (2000) focus on the acquisition of the German particle "auth" ('also'), observing that it appears quite early in development, already at the two-word utterance stage. While they focus on "auth", they also note that "wieder" ('again') is structurally similar and appears early in development.

[5] Materials were slightly different for children above and below 2;06: the older group watched pre-recorded videos, while the younger group watched the experimenters act out the events with toys.

series of short stories, each involving two characters, only one of whom satisfied the presupposition of "again". For example, in one story, a dinosaur crawled under a tree twice, while a crocodile did so only once. After hearing the story, participants were asked to judge a puppet's utterance, which was either about the character who had repeated the action, making the use of "again" felicitous ("the dinosaur crawled under the tree again"), or about the character who only performed the action once, making the use of "again" infelicitous ("the crocodile crawled under the tree again"). Preschool-aged children successfully accepted the felicitous sentences and rejected the infelicitous ones. When asked to explain their rejection of the puppets' infelicitous sentences, they correctly pointed out the reason (e.g., the crocodile crawled under the tree only once/the crocodile had not crawled under the tree before), which suggests that they correctly understood the repetitive reading of "again". However, this finding was based on a subset of the collected data, as up to 40% of child participants were excluded for seemingly ignoring "again" – that is, they accepted (almost) all of the test sentences containing "again".[6] The authors speculated that because the assertion of these test sentences matched the given context, children might have downgraded the relevance of presupposition and prioritized the true assertion. It is also possible that children noticed the presupposition failure, but were more pragmatically tolerant (see Davies & Katsos, 2010; Katsos & Smith, 2010) and thus accepted the test sentences, even though they recognized that they were not perfect descriptions. Ultimately, the authors acknowledge that since presupposition failures do not clearly correspond to a 'false'/'no' response, the TVJT might not be the most effective method for accessing children's knowledge of presupposition.

Because sentences involving presupposition failure are not necessarily judged as 'false' by adults, as discussed in Section 1, the binary tasks employed in previous studies, including the interactive play-based task and the TVJT, may be less than optimal for accessing children's knowledge of presupposition of "again". To address this methodological concern, the present study investigates Mandarin-acquiring children's comprehension of the presupposition of "you" (2), using alternative tasks that, importantly, avoid asking participants to make binary judgments ('true'/'false' or 'yes'/'no') about sentences involving presupposition failure.

2.2. Previous methods for studying presuppositional particles

The literature contains several alternative methods designed to probe children's knowledge of presupposition. In this section, we consider three such methods that have been used in previous studies investigating children's comprehension of "also" and "too". Like "again", these triggers require an additional contextually relevant event, and thus may provide methodological inspiration for investigating children's knowledge of "again".

One such method is a reward-based task developed by Berge and Höhle (2012), who investigated German-speaking children's comprehension of "auch" ('also'). In this task, participants were asked to decide whether to reward a puppet under the rule that it could only be rewarded if it performed two tasks. The puppet was out of sight while performing a task (e.g.,

[6] Similar response patterns have been observed in other studies. For example, in the TVJT of Liu (2009, 2015) and Liu et al. (2011), one child among the nine participants consistently accepted test sentences containing "you" in cases of presupposition failure, explaining that "the little panda ate a banana" (when the panda had eaten a banana only once). Studies on the additive particles "also" and "too" (across languages), conducted using TVJT or picture-selection tasks that require participants to make binary judgments about cases of presupposition failure, also suggest that a sizable number of children appear untroubled by presupposition failures (e.g., German "auch": Hüttner et al., 2004; Dutch "ook": Bergsma, 2002, 2006; Japanese "mo": Matsuoka, 2004; Matsuoka et al., 2006).

eating something). The experimenter would then make a statement about what the puppet must have done, for example, "Oh, there is the lion again! Lion! You've surely eaten the banana!" The puppet's response to the experimenter was either additive, containing the particle "auch" ('also'), e.g., "Guess what? I've also eaten the apple", or corrective, without the particle, e.g., "Guess what? I've eaten the apple." Since the puppet had been out of sight while it carried out the action, the child could only make their decision (to reward the puppet or not) based on what the puppet said (and in particular, whether the puppet used the additive particle). The results showed that three- and four-year-olds were more likely to reward the puppet after hearing sentences containing "auch" than after hearing sentences without it.

Another method we considered was the listener identification task developed by Aravind (2018) and Aravind et al. (2023), which requires participants to identify the intended listener based on what information is in the common ground. Using their experiment on "too" to illustrate the task, children heard stories in which a character (Hippo) ate food (e.g., a donut) in front of one friend (Cat) but not another (Fox). After the friend left, Hippo then ate something else (e.g., a cupcake). Afterwards, Hippo reported what happened to a visitor by producing either an utterance containing "too" (e.g., "Guess what, I ate a cupcake, too, today!") or an utterance without "too" (e.g., "Guess what, I ate a donut today!"). The results suggest that 4–6-year-olds successfully tracked presupposed information and integrated it into their reasoning about likely addressees. They consistently chose the addressee who had witnessed the presupposed event after hearing an utterance with "too", and rejected this addressee after hearing an utterance without "too".

A third task we considered was a Question-Answer Task developed by Kurokami et al. (2021, 2022) and Kurokami (2025), who examined preschoolers' comprehension of English "also"/"too" and Japanese "mo". Children listened to short stories in which three characters each performed one or two actions. At the end of each story, a puppet asked a *wh*-question with or without the particle (e.g., "Who (also) ate a banana?"). The task is based on the idea that presupposed information, in addition to at-issue content, restricts the set of appropriate answers to a *wh*-question. For example, in response to "Who also ate a banana?" (in the kinds of contexts tested in Kurokami et al.), an appropriate answer should name someone who not only ate a banana, but also performed another relevant action. In contrast, naming the individuals who ate a banana (whether or not they did something else) would be appropriate for the question without the particle. The results showed that children were more likely to select the character who had performed both actions when the particle was present than when it was absent, indicating sensitivity to the additive presupposition.

Each of these methods provides useful insights into how to probe children's knowledge of presupposition. Comparing these three methods, we opted to employ a Question-Answer Task most similar to that of Kurokami et al.'s, as our first step to examine children's comprehension of "you". Although results from the reward-based task showed that children generally rewarded the puppet after hearing sentences containing "also", non-adultlike responses persisted for sentences without the particle (the reward rate was 63.5% for 3-year-olds and 38.2% for 4-year-olds).[7] This highlights a potential concern that children may have a general tendency to reward the puppet regardless of the presence or absence of the particle (see Kurokami et al., 2021; Kurokami, 2025). The listener identification task, on the other hand, places additional demands on children's Theory of Mind (ToM), as participants must reason about Hippo's beliefs about the addressee's knowledge,

[7] Similar non-adultlike behavior was reported in Huang et al.'s (2025) study on Mandarin *ye* 'also', though at lower rates.

in addition to interpreting the presupposition trigger.[8] Comparatively, the Question-Answer task more directly targets children's sensitivity to the presuppositional content, while imposing fewer extra-linguistic demands.

How does the Question-Answer task help us to understand children's understanding of "again"? Like the presuppositions of the additive particles "too" and "also", the presupposition of "again" can restrict the set of appropriate answers to a *wh*-question, in addition to the at-issue content. For example, in response to a question such as "Who sneezed again after the sun set?", an appropriate answer must name someone who sneezed both after the sunset and earlier. In contrast, naming the individual(s) who sneezed after the sunset (whether or not they sneezed earlier) would answer the question without "again". As detailed below, in our task, participants had to answer a *wh*-question about a story that had unfolded. We set up the scenarios in such a way that two characters would differ in whether they satisfied the presupposition of "you". In this way, the participant was expected to be able to correctly answer a question containing "you" only if they understood that the question presupposed the earlier action.

3. Experiment 1: Question-Answer Task

3.1. Methods

3.1.1. Participants

40 Mandarin-acquiring children (one 3-year-old, 23 4-year-olds and 16 5-year-olds ranging from 3;11 to 5;11, mean age: 4;10, *SD* = 0.52, 20 girls) and 34 adult native speakers of Mandarin participated in the experiment.[9] An additional 11 children were tested but excluded because they failed to meet our inclusion criterion of scoring at least 75% accuracy on fillers, and an additional five children were excluded because a parent interfered with the session.

Both adult and child participants were recruited and tested individually online (via Tencent Meeting) during the COVID-19 pandemic.

3.1.2. Procedure

Participants listened to a series of pre-recorded, illustrated short stories, each featuring two contrasting characters engaging in some activity before and after the sun set. We reasoned that sunset was a time point for which it was relatively easy to visually depict a 'before' and 'after' period; moreover, children should be familiar with the concept of sunset (as opposed to, for example, a particular time or a particular day of the week).

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[8] The Theory of Mind abilities involved in the listener identification task, which requires participants to reason about someone's belief regarding another individual's knowledge, are arguably more complex than those involved in traditional ToM tasks, which typically require participants to reason about an individual's mental state in a non-recursive manner. It is worth noting, however, that previous research on the acquisition of "too" (Aravind, 2018; Aravind et al., 2023) has shown that preschool children can successfully complete this task and perform in an adult-like manner. Future research could therefore explore the use of this paradigm to investigate children's knowledge of other presupposition triggers, including "again".

[9] Production data show that children begin using "again" felicitously to denote repetition before the age of 3;06 (Lee, 2005; Penner et al., 2000; Xu & Snyder, 2017; Xu et al., 2024). We chose to conduct our experiment with slightly older children (above 3;06) to align with previous experimental studies of children's comprehension of "again" and "you", and also to align with Kurokami et al. (2021, 2022) and Kurokami (2025), who used the same methodology.

Between the two characters in each story, only one satisfied the presupposition of “you”, by carrying out an action twice. At the end of each story, participants were presented with a pre-recorded *wh*-question, which asked who had performed the relevant action. Participants had to choose between the character who had done the action once, the character who had done the action twice, and a ‘both’ option if they felt that both characters had done the relevant action. We provide a translated example of a target trial in (5) (paired with Figure 1), but keep the critical *wh*-question in Mandarin. In the *wh*-questions on the target trials, “you” was stressed in a natural way to make the adverb more salient.

(5) *Story:* Today Cat and Fox were playing with some very ticklish magical feathers in the forest. They made up a game: even if a feather tickles you, you can’t sneeze! Cat tickled Fox’s nose with a feather. Fox let out a big sneeze, “Ahhh-choo!” Then Fox tickled Cat’s nose with a feather. Cat managed not to sneeze. Soon, the sun set and it was evening. The two friends decided to play one more round. Cat found a feather and tickled Fox’s nose. Fox let out a big sneeze, “Ahhh-choo!” Then Fox tickled Cat’s nose with a feather. Cat let out a big sneeze, “Ahhh-choo!”

Experimenter: We know Cat and Fox played with feathers before and after the sun set.

Taiyang	luo-shan	hou	shui	**you**	da-le-penti:
Sun	fall-mountain	after	who	again	hit-ASP-sneeze
huli,	xiao-mao,	haishi	taimen	liang-ge?	
fox	little-cat	or	they	two-CL	

‘Who sneezed again after the sun set: Fox, Cat, or both?’

Target response: Fox

Figure 1: Pictures paired with example (5), depicting the outcome of the sneezing game before (left) and after (right) the sun set. Both the pre-sunset and post-sunset scenes were displayed at the end of the trial, in order to avoid memory issues (i.e., to remind participants of the sequence of events).

3.1.3. Materials

We tested four different predicates: “da-penti” (‘sneeze’), “shuai-dao” (‘fall’), “shui-zhao” (‘fall asleep’), and “pao-bu” (‘run’). The target trial with the predicate ‘sneeze’ is presented in (5). If a participant was sensitive to the presupposition of *you*, they were expected to choose the character who performed the action twice. The expected answer for (5) was thus Fox, as he had sneezed earlier, before the sun set. Conversely, if participants were not sensitive to the presupposition of "you" or completely ignored it, they were expected to select the 'both characters' option.

For each of the predicates that appeared in the target condition, we also included a control trial in which the *wh*-question did not contain "you" ('again') (see example (6), paired with Figure 2). Participants were asked "Who sneezed after the sun set?" In this case (in the absence of "you"), the expected answer was 'both' characters.

(6) *Story:* Today Giraffe and Duck were tidying the kitchen, which was very dusty. They each moved a box from the corner to the middle of the kitchen. Duck accidentally dropped his box on the floor, and there was a big cloud of dust. Duck let out a big sneeze, "Ahhh-choo!" Giraffe gently set his box down and managed not to sneeze. Soon, the sun set and it was evening. The friends decided to move the last boxes. Duck tried to set his box down gently, but still there was a cloud of dust. Duck let out a big sneeze, "Ahhh-choo!" Then Giraffe accidentally dropped his box on the floor, and there was a big cloud of dust. Giraffe let out a big sneeze, "Ahhh-choo!"

Experimenter: We know Duck and Giraffe moved some boxes before and after the sun set.

Taiyang	luo-shan	hou	shui	da-le-penti:
Sun	fall-mountain	after	who	hit-ASP-sneeze
Xiao-ya,	changjinglu,	haishi	taimen	liang-ge?
little-duck	giraffe	or	they	two-CL

'Who sneezed after the sun set: Duck, Giraffe or both?'

Target response: Both

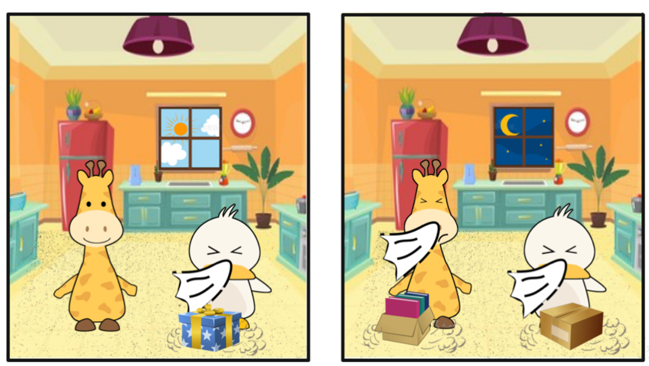


Figure 2: Pictures paired with example (6), depicting the outcome of moving boxes before (left) and after (right) the sun set.

All participants received two training items in the same order, followed by 16 trials (four critical "you" targets, four controls without "you", and eight fillers) in a blocked design, with the targets preceding the controls. This blocked order was chosen to ensure that the absence of "you" on control trials would not influence participants' interpretation of the target trials. Without a block design, the critical targets and controls would appear in a randomized order. If a participant encountered a control trial without "you" first, this could encourage them to (continue to) respond in a manner consistent with the absence of "you". Four filler items were interspersed within each block, leading to a total of 8 trials in each block.

The stories and test questions were presented in the form of pre-recorded videos. The training and filler items were simple stories, each involving two characters engaging in activities, followed by a *wh*-question at the end asking who had done something. For instance, in the story where Chicken bought a guitar and a violin, and Sheep bought a piano and a violin, participants were asked "Who bought a guitar – Chicken, Sheep or both?" Among the eight filler items, two had 'both' as the target answer, and the other six had one of the characters as the target answer. This ensured that the three possible answers (the first character in the *wh*-question, the second character in the *wh*-question, and 'both' characters) were roughly equal in number.

We counterbalanced whether the expected target character appeared to the left or the right, to control for potential preferences for a particular side. Trials were fully randomized across participants within the target and control blocks using a Python script.

3.2. Results

Figure 3 displays the proportion of selections of the 'repeater' character, i.e., the character who performed the action twice, in the control and target conditions, across children and adults. We can see from the figure that adults, as expected, tended to select the repeater in the target condition, but not in the control condition. Children's behavior did not mirror that of adults: while they tended not to select the repeater in the control condition, they did not show a tendency to select it in the target condition either.

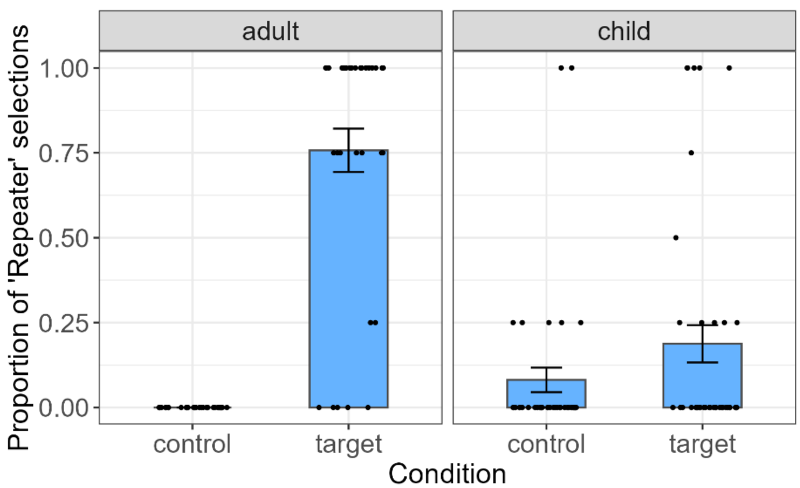


Figure 3: Proportion of selections of the 'repeater' character in control and target conditions, across children and adults. Error bars represent the standard error of the mean, and individual dots indicate each participant's mean proportion of selecting the repeater character.

None of the adult participants selected the repeater character in the control condition. The lack of variance in this condition precluded the use of ANOVAs or linear models for analyzing the full dataset. We thus conducted two Mann-Whitney U tests on the target and control conditions, respectively. The results for the target condition indicate that children selected the repeater character significantly less than adults did ($W = 1122.5$, $p < .001$, Bonferroni-adjusted for multiple comparisons). For the control condition, the proportion of selecting the repeater differed significantly across the two participant groups (*W*=561, *p*=.011, Bonferroni-adjusted for multiple comparisons). The 95% confidence interval for the proportion of children selecting the repeater character in the control condition (0.01 – 0.16) did not cross 0, suggesting the same pattern.

Figure 4 plots the proportions of each response type given by children and adults in the target and control conditions. The plot shows that when adults and children did not select the repeater character in the target condition, the majority of them opted for the 'both' option. This would be the expected answer if the question did not contain "you" ('again'). In such cases, the question essentially inquires about what happened after the sun set, and in all cases, both characters performed the action after the sun set.

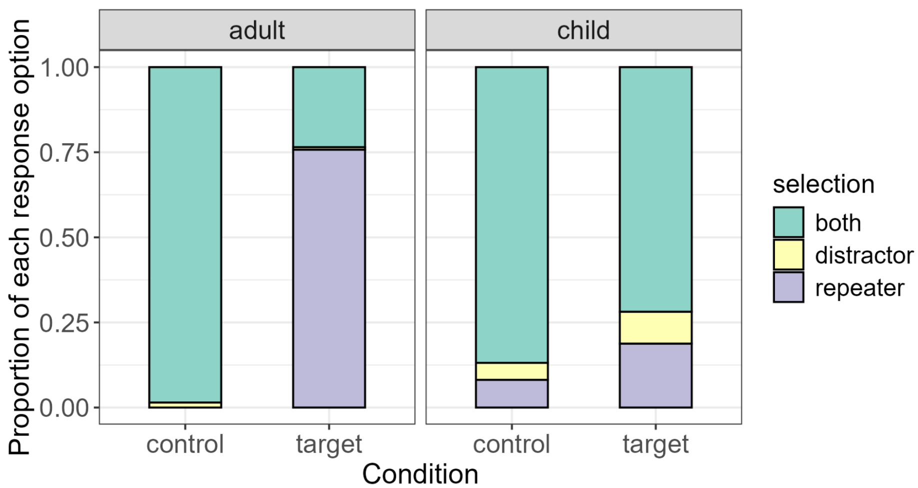


Figure 4: Proportion of selections of each response option.

To better understand children's selections in the target and control conditions, we next analyzed the child data using multinomial logistic regression, using the mclogit package (Elff, 2022) in R version 4.3.1 (R Core Team, 2021).[10] For the three-level outcome variable (both, distractor, repeater), we set the response 'both' as the reference category. We considered models with varying random effect structures, including models with and without random intercepts and random slopes for Subject and Item. For fixed effects selections, we used the best subset regression method (Miller, 2002), considering models with all possible combinations of the following predictors: Condition (target vs. control), Age (as a continuous variable in years and centered) and their interaction. Model fit was evaluated based on Akaike Information Criterion (AIC), following Bolker et al. (2009). The model with the lowest AIC included Condition and Age as predictors, and a random intercept for Subject (Selection ~ 1 + Condition + Age + (1 | Subject)). The model results are summarized in Table 1. For 'repeater' responses (vs. 'both'), there was a significant effect of Condition: children were more likely to select the repeater in the target condition than in the control condition. The effect of Age was not significant. For 'distractor' responses (vs. 'both'), there was again a significant effect of Condition: children were more likely to choose the 'distractor' in the target condition than in the control condition. The effect of Age did not reach significance, but the direction of the effect suggests a trend for younger children to give more distractor responses (see also the Appendix).

[10] We did not fit multinomial logistic regression models to the adult or combined data, because no adult participants selected the repeater response in the control condition.

Table 1. Results of multinomial logistic regression model. The outcome variable was coded with ‘both’ as the reference level.

	β	*SE*	*z*	*p*	OR [95% CI]
repeater vs. both					
Condition (target)	2.36	0.58	4.10	<.001	10.55 [3.42, 32.56]
Age (centered)	0.21	0.99	0.21	.834	1.23 [0.18, 8.50]
distractor vs. both					
Condition (target)	1.99	0.69	2.89	.004	7.28 [1.89, 28.00]
Age (centered)	-2.06	1.11	-1.85	.064	0.13 [0.02, 1.13]

3.3. Discussion

Experiment 1 used a Question-Answer Task designed to avoid asking participants to make binary judgments (e.g., ‘true’/‘false’, ‘yes’/‘no’) about sentences exhibiting presupposition failure. However, the results ran counter to the previous studies: while children exhibited some sensitivity to “you”, their performance was not adult-like. Most of them selected the ‘both’ option, as if ignoring the presupposition trigger.[11] Even among adults though, we observed some proportion of selecting the ‘both’ option.

Let us consider first why adults might have opted for the ‘both’ option. One possibility may be related to the so-called ‘agentless’ presupposition reading of “again”, which requires only that the action be repeated, regardless of who performs it (see Asami & Bruening, 2025; Ausensi et al., 2021; Bale, 2007; Smith & Yu, 2022, 2024; Zhang, 2022). This reading is illustrated in (7), where there are two events of guessing the riddle wrong, but crucially each involving a different agent – first the big brother, and then the little brother.

(7)

Gege	cai-cuo-le,	didi	you	cai-cuo-le.
Big.brother	guess-wrong-ASP	little.brother	again	guess-wrong-ASP

Zhe-ge	miyu	ke	nan	le.
this-CL	riddle	rather	difficult	SFP

‘The big brother guessed the riddle wrong, and so did the little brother. This riddle is quite difficult.’

(Zhang, 2022, p. 88, ex. (12a))

To illustrate how this reading might have been at play in the experiment, consider the example story in (5), where two sneezing events took place: only Fox sneezed before the sun set, but both

[11] Importantly, this tendency cannot be explained by children quickly deciding on their answer before they have even heard the presupposition trigger, since in Mandarin “you” is preverbal (J. Lidz, personal communication, November 4, 2023).

Cat and Fox sneezed after the sun set (hence Fox sneezed twice, while Cat sneezed only once). This story was followed by a question about who sneezed again after the sunset. Under the agentless presupposition reading, the *wh*-question presupposes that a prior sneezing event occurred before the sunset, which is clearly satisfied in the story (by Fox's sneezing before the sunset), and asks about the agent of the sneezing event that took place after the sunset. Some of the adult (and child) participants may have accessed the agentless presupposition reading, leading them to respond 'both'. That there were relatively few 'both' selections in adults suggests this reading is not very salient, or at least is not the predominant reading; on the other hand, if this reading is indeed responsible for the 'both' response, it seems that children were much more willing to respond on the basis of this reading, compared to adults.

Why might children have selected 'both' more often than adults? Here we identify a few different possible explanations for the results, noting that these explanations are not necessarily mutually exclusive, and could in fact all play a partial role in accounting for the results. First, setting aside the possibility of an agentless presupposition reading, children at this age might simply lack the relevant presuppositional knowledge of "you", causing them to disregard the trigger.

Another possibility is that the presupposition of "you" was not made sufficiently salient in the experimental contexts. Although "you" was stressed in the *wh*-question, it might not have been prominent enough for child participants to pay attention to it. Even without considering the trigger, they would still have been able to arrive at an interpretation and provide an answer.

Yet another possibility centers on participants' ability to project the presupposition from the *wh*-question. It has been suggested that presuppositions in *wh*-interrogatives project universally (Abrusán, 2014; Nicolae, 2015; Schlenker, 2008). According to this view, a *wh*-question such as (8) would presuppose that all the individuals in the domain (i.e., Cat and Fox) sneezed before.

(8) Who sneezed again after the sun set: Cat, Fox, or both?
Presupposition: All individuals in the domain (Cat, Fox) sneezed before.

Meanwhile, certain pragmatically non-canonical questions, such as quiz questions, consistently exhibit projection patterns weaker than universal projection (Schwarz & Simonenko, 2018; Theiler, 2020). To characterize the contrast between canonical information-seeking questions and non-canonical questions in their presupposition projection patterns, Theiler (2020) proposes what she calls the "Epistemic Bridge", as described in (9).

(9) A question Q is felicitous relative to a context set c only if the presuppositions of all answers that the speaker considers possibly true are entailed by the context set.
(Theiler, 2020, ex. (22))

The intuition behind (9) is that the speaker of a question wants to avoid a situation in which the desired common ground update cannot take place because a true answer turns out to be infelicitous. For a canonical information-seeking question, because the speaker is agnostic about the possible true answers, universal projection of the presupposition would ensure that all the true answers are also felicitous. In contrast, for non-canonical information-seeking questions, the speaker already has prior knowledge about which questions will turn out to be true. As a result, the question allows for (weaker) existential projection of embedded presuppositions.

In our experiment, the test questions containing "you" are not canonical information-seeking questions. The speaker (the experimenter) already knows which answer is true and felicitous. In other words, one can draw a prior knowledge implicature. Adult participants, as addressees, can make this prior knowledge inference and restrict the domain of answers to those the speakers considers possibly true. Unlike adults, however, children may project the presupposition differently from *wh*-questions, instead adhering to a universal-projection interpretation, which leads to presupposition failure (e.g., in (5), it is false that both Cat and Fox sneezed earlier). This could consequently lead children to ignore "again" and select 'both'. As for why children might favour a universal-projection interpretation, there could be multiple reasons. Children may have difficulty restricting the domain of possible answers to those the speaker (the experimenter) considers possibly true; or they may have difficulty identifying what knowledge is possessed by the speaker in the experimental context, which has to do with their ToM abilities; or they may have difficulty distinguishing canonical and non-canonical information-seeking questions.[12]

In Experiment 2, we moved to a Felicity Judgment Task, which improved upon Experiment 1 by ensuring sufficient salience of the trigger by using direct contrast to highlight the presence of "you".[13] The second experiment also circumvented the issue of presupposition projection by testing the interpretation of "you" in declarative sentences instead of *wh*-questions, all the while not requiring participants to judge the truth or falsity of a sentence with presupposition failure. Additionally, in order to assess the felicity of "you", participants had to determine whether a mentioned character had repeated the action or not, making it less likely for them to access the agentless presupposition reading.

4. Experiment 2: Felicity Judgment Task

Experiment 2 builds on the observation that presupposition triggers like the additive particle "too", the iterative particle "again", and the definite determiner "the", are obligatory when their presuppositions are satisfied in the context (see Chemla, 2008; Percus, 2006; Sauerland, 2008 for English; Bade, 2016; Heim, 1991 for German; Amsili et al., 2016 for French). This is illustrated in (10), where the absence of English "again" results in infelicity. A similar pattern is observed for the Mandarin translational equivalent "you", as shown in (11).

(10) Jenna went ice skating yesterday. She went ice skating today, #(again).
(Bade & Renans, 2021, ex. (1b))

[12] We thank an anonymous reviewer for pointing out some of these possibilities.

[13] An anonymous reviewer suggests a possible follow-up experiment in which universal presupposition projection in *wh*-questions does not lead to presupposition failure. We explored this option in a pilot experiment with adults in which we embedded the question-answer pair in a larger discourse context by first asking, "Who sneezed before the sunset?" The answer to this question was intended to serve two purposes: (1) to provide a salient antecedent for "you", and (2) to restrict the domain of the answer to the 'after'-question to individuals who satisfied the presupposition of "you". However, we found that adult participants in this pilot study selected 'both' at an even higher rate. We speculate that the explicit 'before' and 'after' framing of the question pair induces a strong contrastive focus, making the agentless presupposition reading (i.e., that the event represented by the predicate happened before) more salient. Adopting this design for child participants would therefore risk further encouraging children to access this reading, potentially increasing the rate of 'both' responses. We therefore adopted a different method in Experiment 2 to investigate children's sensitivity to the presupposition trigger "you".

(11) Zhangsan zuotian qu-huabing le. Jintian #(you) qu-huabing le.
Zhangsan yesterday go-skate ASP today again go-skate ASP
'Zhangsan went skating yesterday. He went skating today, #(again).'

We will return to examples like (10) and (11) in the General Discussion section, after we have presented Experiment 2.

4.1. Methods

4.1.1. Participants

43 Mandarin-acquiring preschool-aged children (30 4-year-olds and 13 5-year-olds, age range: 4;0–5;11, M=4;10, *SD* = 0.50, 18 girls) and 33 adult native speakers of Mandarin participated in the experiment.[14] An additional six children were excluded because they failed to meet our inclusion criterion of scoring at least 75% accuracy on fillers.

4.1.2. Procedure

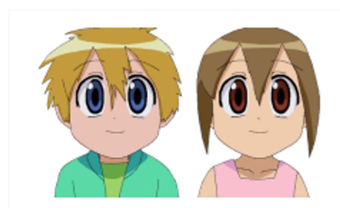
We implemented a Felicity Judgment Task in which participants watched the same set of stories as those in Experiment 1. After each story, the experimenter asked a question and two speakers (a boy and a girl) answered the question. Participants were asked to determine who had answered better; see example (12).[15,16]

(12) *Story*: [same as in (5)]

Experimenter: We know Cat and Fox played with feathers before and after the sun set. What happened after the sun set?

Boy: Huli **you** da-le-penti.
Fox again hit-ASP-sneeze
'Fox sneezed again.'

Girl: Huli da-le-penti.
Fox hit-ASP-sneeze
'Fox sneezed.'



Experimenter: Who answered better?

Target response: Boy

[14] None of the participants had completed Experiment 1.

[15] An anonymous reviewer highlights that although the Felicity Judgment Task effectively accomplishes what we are after by highlighting the contrast between the presence and absence of the presupposition trigger, this task introduces an element of competition that is absent from the Question-Answer Task used in Experiment 1. It is worth noting a similar element of competition in Experiment 1, however, in that in order to answer the test questions, participants had to draw a contrast between the character who had done the action once and the character who had done it twice. In any case, we agree with the reviewer that the 'competitive' nature of the Felicity Judgment Task makes it different from the Question-Answer Task, and that such differences should not be treated as trivial.

[16] As in Experiment 1, both the pre-sunset and post-sunset scenes were displayed at the end of each story, in order to help participants keep track of what happened in each story.

The experiment was implemented using Qualtrics. The stories and the two speakers' responses were pre-recorded. The adult participants completed the experiment independently via a provided link, where they could read the question after each story. The child participants were tested individually by the experimenter, who read the question aloud after each story.

4.1.3. Materials

We had both a target and a control condition in the experiment; in both conditions, we tested the same four predicates as in Experiment 1: "da-penti" ('sneeze'), "shuai-dao" ('fall'), "shui-zhao" ('fall asleep'), and "pao-bu" ('run').

As illustrated in (12), for the target trials, the boy and the girl each answered the experimenter's question by mentioning the character who had repeated the action. One speaker used "you" ('again') while the other did not. The expected response on these trials was the speaker who had used the trigger "you" ('again'), because its presupposition was satisfied, thus making its presence obligatory.

In the control condition, both speakers answered the experimenter's question by mentioning the character who had performed the action only once. The expected response was to select the speaker who had not used "you" ('again'), because the utterance containing "you" would involve a presupposition failure; see example (13).

(13) *Story:* [same as in (6)]

Experimenter: We know Duck and Giraffe moved some boxes before and after the sun set. What happened after the sun set?

Boy:	Changjinglu	**you**	da-le-penti.
	Giraffe	again	hit-ASP-sneeze
	'Giraffe sneezed again.'		
Girl:	Changjinglu	da-le-penti.	
	Giraffe	hit-ASP-sneeze	
	'Giraffe sneezed.'		

Experimenter: Who answered better?

Target response: Girl

All participants received two training items, followed by 16 trials: four targets, four controls, and eight fillers, presented in a randomized order. We counterbalanced which speaker corresponded to the target response and which speaker responded first, to control for any potential preferences children might have for one of the two speakers.

In the filler items, the two speakers also answered a question by mentioning a character, with their responses differing in truth value. For example, in a story in which Chicken bought a guitar and a violin, and Sheep bought a piano and a violin, the experimenter asked, "We know Chicken and Sheep each bought a violin. What else did they buy? The boy responded, "Sheep bought a piano" (true) while the girl responded, "Sheep bought a guitar" (false). Participants were then asked to choose who had answered better.

4.2. Results

Figure 5 displays the proportion of responses selecting the speaker who used “you” (‘again’), across control and target conditions, in children and adults.

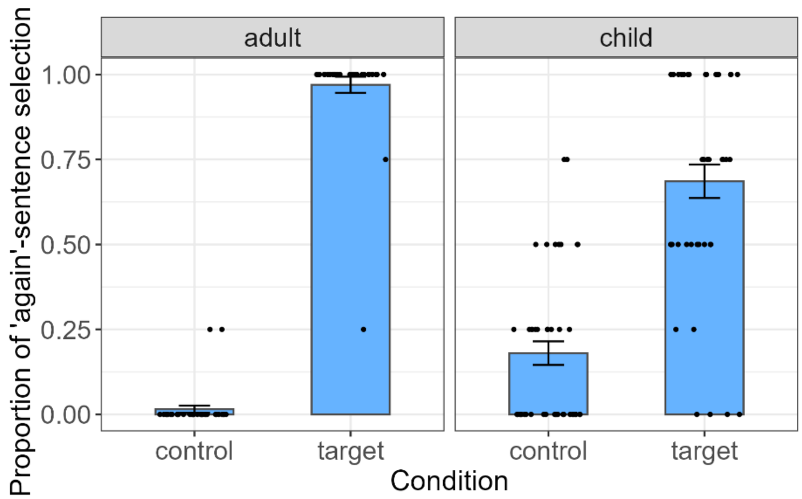


Figure 5: Proportion of responses selecting the speaker who uttered “you” (‘again’) in control and target conditions, across adult and child participants. Error bars represent the standard error of the mean, and individual dots indicate each participant’s mean proportion of selecting the speaker who used “you”.

We analyzed the data with a mixed effect logistic regression model with the lme4 package (Bates et al., 2015) in R. The dependent variable was participants’ responses (1=selecting the again-speaker, 0=not selecting the again-speaker). Fixed effects included Condition (target vs. control), ParticipantGroup (adult vs. child), and their interaction. Following Barr et al. (2013), we started with a model with a maximal random effect structure (including random slopes and intercepts for Subject and Item), and then reduced the model’s complexity stepwise until it converged to a non-singular fit. This resulted in a model that included by-participant random slopes and intercepts. Likelihood ratio tests indicated that adding the interaction significantly improved model fit ($\chi2(1) = 34.49$, $p < .001$). We therefore included all three fixed-effect variables in the final model: Selection ~ 1 + Condition * ParticipantGroup + (1 + Condition | Subject). Model results revealed a significant effect of Condition ($\beta = 9.71$, $SE = 1.29$, $z = 7.55$, $p < .001$) and ParticipantGroup ($\beta = 2.83$, $SE = 0.80$, $z = 3.54$, $p < .001$). In addition, there was a significant interaction between Condition and ParticipantGroup ($\beta = -6.62$, $SE = 1.30$, $z = -5.10$, $p < .001$). These results indicate that both groups significantly favored the speaker who used “you” in target trials compared to control trials, with adults exhibiting a larger difference between targets and controls than children.

Figure 6 displays the different response types given by the different participant groups. The category ‘other’ includes trials where participants either could not give a response or responded by choosing both speakers. Both groups favored the speaker who used “you” in the target condition and the speaker who did not use “you” in the control condition. While children’s responses were noisier than those of adults, their preferred responses mirrored those of adults.

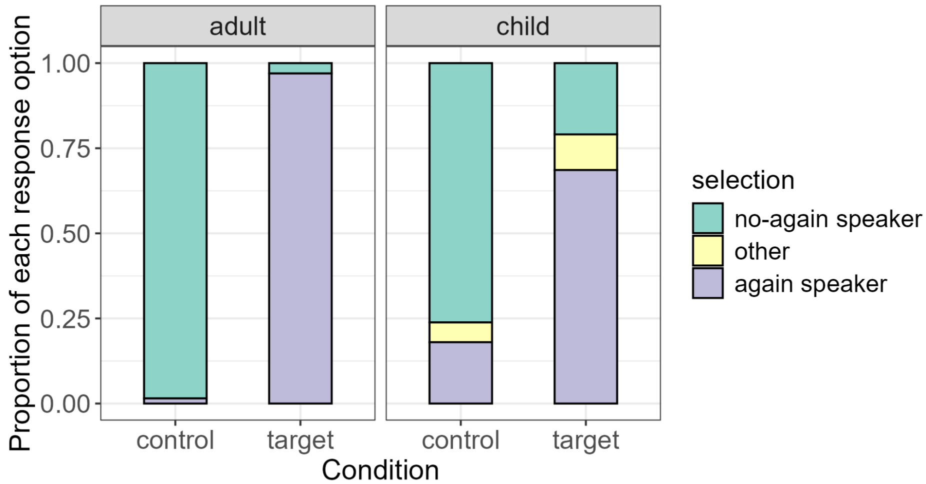


Figure 6: Proportion of selections of each response option. 'Other' responses corresponded to trials where participants could not give a response, or answered by choosing both speakers.

To better understand children's selections in the target and control conditions, we next analyzed the child data using multinomial logistic regression. For the three-level outcome variable (selecting the speaker with "you", selecting the speaker without "you", and 'other'), we set the response of selecting the speaker with "you" as the reference level. We considered models with varying random effect structures, including models with and without random intercepts and slopes for Subject and Item. We also considered fixed effect models with all possible combinations of the following predictors: Age (as a continuous variable and centered), Condition (target vs. control), and their interaction. The model with the lowest AIC included all predictors and a random intercept of Subject (Selection ~ 1 + Condition * Age + (1 | Subject)). The model results are summarized in Table 2. For the contrast between 'no-again speaker' vs. 'again-speaker', there was a significant effect of Condition, suggesting that children were overall more likely to choose the speaker who used *you* in the target trials compared to control trials. There was also a significant effect of Age, suggesting that higher age increased the odds of choosing the speaker who did not produce "you" (generally related to improved performance on the controls, see Appendix). Finally, the interaction between Condition and Age was significant, indicating that the effect of Condition was more pronounced amongst older children. For the contrast between 'other' responses vs. again-speaker responses, on the other hand, there were no significant effects of Condition, Age, or their interaction.

Table 2. Results of the multinomial logistic regression model. The outcome variable was coded with the '*again*-speaker' as the reference level.

	β	*SE*	*z*	*p*	OR [95% CI]
no-*again*-speaker vs. *again*-speaker					
Condition (target)	-3.04	0.32	-9.41	<.001	0.05 [0.03, 0.09]
Age (centered)	1.85	0.66	2.80	.005	6.36 [1.74, 23.17]

Age: Condition	-2.63	0.73	-3.62	<.001	0.07 [0.02, 0.30]
other vs. *again*-speaker					
Condition (target)	-0.57	0.52	-1.11	.265	0.56 [0.21, 1.55]
Age (centered)	1.47	1.11	1.32	0.187	4.34 [0.49, 38.42]
Age: Condition	-2.01	1.20	-1.68	0.093	0.13 [0.01, 1.40]

4.3. Interim summary

Experiment 2 shows that children successfully understand the presuppositional trigger "you" ('again'), though not yet at fully adult-like levels. As in Experiment 1, the task avoids the problem of forcing participants to make binary judgments about sentences exhibiting presupposition failure, and unlike previous studies, we did not have to exclude children who appeared to ignore the trigger. Furthermore, this method set aside some of the issues we identified as potentially leading to children's selection of the 'both' response in Experiment 1. First, it resolved the issue of insufficient saliency of the trigger: the only difference between the two speakers' descriptions was the presence/absence of "you". If participants ignored the trigger, they would be comparing two identical sentences. Second, since both speakers focused on one animal character when responding to the experimenter's questions, participants needed to consider whether the mentioned character had repeated the action in order to assess whether the use of "you" was felicitous. This setup made it less likely for participants to access the agentless presupposition reading (according to which it does not matter who repeated the action). Moreover, since participants were asked to compare two declarative sentences, the task overcame the issue of children's potential difficulty with presupposition projection from *wh*-questions.

5. General Discussion

In this paper, we conducted two experiments to tap into Mandarin-speaking children's knowledge of the presupposition trigger "you" ('again'). The results of the Question-Answer Task suggest children were simply ignoring the presupposition trigger, while the results of the Felicity Judgment Task suggest sensitivity to the presupposition (albeit not at an adult-like level). In this section, we discuss the broader implications of our findings and highlight some questions for future research. We begin by discussing the results of the Felicity Judgment Task, focusing on children's knowledge of the presupposition of "you". Next, we consider these results in the context of children's knowledge of the obligatory presence of a trigger when its presupposition is satisfied. We then explore in greater depth two underlying factors that may have contributed to children's selection of the 'both' option in the Question-Answer Task: their access to the agentless presupposition reading and their ability to project presuppositions from *wh*-questions.

5.1. Children's knowledge of "you": a possible production-comprehension asymmetry

As reviewed in Section 2, previous studies of children's production have shown that they are able to correctly produce "you" ('again') at a fairly early age (before 3;06). In our Felicity Judgment

Task, however, although 4- and 5-year-old children displayed sensitivity to the presupposition of "you", their performance was still not fully adult-like. These findings suggest that children's adult-like production of the presupposition trigger may precede their comprehension of it. However, the finding that children's comprehension is not fully adult-like may not be entirely due to a lack of knowledge of "you", but instead could be related to processing demands of the relevant tasks. The Felicity Judgment Task requires participants to hold both speakers' responses in memory, and compare and evaluate them against the story, which may impose a substantial processing load on preschoolers. To further probe their knowledge, we examined individual response types in the control condition, which tested whether children could recognize presupposition failure and subsequently select the statement without "you". We defined participants as responding consistently if they selected the same option on at least three out of four trials. By this measure, 30 out of 43 children consistently chose the "again"-less statement, 2 chose the again-speaker, 1 chose 'other', and the remaining 10 responded inconsistently. These results suggest that most children recognized the presupposition failure of "you", and their non-adult-like performance likely reflects processing demands rather than a lack of presuppositional knowledge of the trigger.

The individual response patterns also suggest that children are not simply interpreting "you" as an expression like "more than once"/"twice". If they were, we would expect participants in both target and control conditions to consistently choose the speaker who did not use "you", since none of the characters in the stories performed the action more than once after the sunset. In fact, only one child chose the 'again'-less speaker on at least three out of four trials across both conditions, contrary to this prediction. This raises a deeper question about how children come to identify 'again' as a presupposition trigger, and more broadly, how learners come to successfully distinguish presupposed information from what is asserted. To our knowledge, this question has not been resolved (but see Tieu et al., 2019; Schlenker, 2021a, b for work that explores such questions using gestures and visual animations; see also Bade et al., 2024 for work on adult word learning that investigates "triggering algorithms", which specify which entailments should be treated as presuppositions). We leave this question for future research.

5.2. On children's knowledge of the obligatory presence of "you" ('again')

The results of the Felicity Judgment Task demonstrate that many preschool children are sensitive to the obligatory presence of "you" when its presupposition is satisfied. In the target contexts, in which the relevant character carried out the action twice, children recognized that it was better to describe the situation using the presupposition trigger "you" ('again') than without it. Here we consider what knowledge might underlie this observed sensitivity.

Several analyses have been proposed to account for the apparent obligatory presence of the trigger, and here we focus on two major formal accounts (see Aravind & Hackl, 2017 for another possibility). The first is *Maximize Presupposition!* (MP), a principle that requires speakers to choose the presuppositionally stronger statement among a set of contextually equivalent expressions (Heim, 1991; see also Amsili & Beyssade, 2010; Chemla, 2008; Percus, 2006; Sauerland, 2008; Schlenker, 2012; Singh, 2011). Initially proposed to account for the contrast between the definite determiner "the" and the indefinite determiner "a", it was subsequently applied to other triggers. For the obligatory presence of "again", the competition lies between a sentence with "again" and one without it. In a context in which the presupposition of "again" is satisfied, MP leads the speaker to utter the sentence containing the trigger.

An alternative account is Obligatory Inference (OI, see Bade, 2016; Bade & Tiemann, 2016;

Bade & Renans, 2021), which argues that the insertion of a trigger is necessary to avoid semantic conflict. Consider (14). In the absence of the trigger, "after the sun set" carries focus, triggering an exhaustive interpretation (i.e., Fox sneezed after the sun set *and at no other time*), which contradicts the preceding sentence. Inserting "again" prevents this contradiction by blocking the exhaustive interpretation, because the presupposition of "again" ensures that a prior event in which Fox sneezed occurred earlier, thus avoiding the conflict.

(14) Fox sneezed before the sun set. He sneezed #(again) [after the sun set]$_F$.

The results of our experiment are in principle compatible with both theoretical accounts (MP and OI), and suggest that knowledge of the relevant mechanism is already emerging in 4- and 5-year-old children. Our experimental study was not designed to tease apart the two theoretical accounts. The findings, however, do raise questions about how child learners might develop the observed sensitivity to the obligatory presence of a presupposition trigger.

Developmental research on this issue has primarily focused on MP, examining different triggers and yielding mixed findings. Some studies suggest early mastery of MP (see, e.g., Wexler, 2003 on children's acquisition of the definite determiner "the" and indefinite determiner "a"), while others point to later mastery (e.g., Yatsushiro, 2008 on German-speaking children's comprehension of the universal quantifier "jeder" ('every') and the definite determiner "der"/"die"/"das" ('the'), particularly in comparison with lexically encoded presuppositions (e.g., Legendre et al., 2011 on 30-month-olds' acquisition of French personal pronouns). Recent investigations further complicate the picture, showing mixed performance for different triggers (see Aravind, 2018 on preschoolers' acquisition of "both" and "another", and Torma et al., 2021 2021 on preschoolers' comprehension of "both" and "either").

Because our results are compatible with both MP and OI, they do not directly determine whether MP is in place early or late. However, if MP indeed governs the obligatory presence of "again", our results suggest that the non-adultlike performance observed in previous studies of other MP-governed triggers stems not from a lack of knowledge of MP, but rather other factors, such as a lack of knowledge about what forms should compete, or a non-adultlike semantic representation of the triggers themselves. On the other hand, no prior developmental work to our knowledge has examined children's knowledge of OI, making it an even less explored area than MP. If instead of MP it is OI that governs the obligatory presence of "you" ('again'), our results could suggest that OI is in place relatively early. Future work could aim to tease apart the two accounts by identifying cases where the two mechanisms make different predictions.

5.3. On children's knowledge of the agentless presupposition interpretation

We proposed earlier that some child (and adult) participants may have selected the 'both' option in the Question-Answer Task because they accessed the agentless presupposition reading of "you" ('again'), on which the relevant action is indeed repeated, but can be performed by a different agent. We speculate that the narrative progression of the stories in our experimental items, which involved two distinct time periods—one before and one after the sun set—may have implicitly encouraged some participants to consider two events and focus on the shift in agents across them, making the agentless presupposition reading more salient.[17]

[17] An anonymous reviewer notes that the Felicity Judgment Task also spanned two time periods (before and after the sunset) and suggests that the agentless presupposition reading might explain children's

Theoretical analyses argue that the agentless presupposition reading is derived structurally, with repetition-denoting 'again' scoping over the predicate while excluding the subject at LF (see Asami & Bruening, 2025; Ausensi et al., 2021; Bale, 2007; Smith & Yu, 2022, 2024; Zhang, 2022). This is illustrated by the Mandarin example (15), with the derivation of the agentless presupposition reading presented in (15c).

(15) a. Lili you qie-le na-zhang da-bing.
Lili again cut-ASP that-CL big-pancake
'Lili cut the big-pancake again.' (ambiguous)

b. repetitive reading: Lili had cut the big pancake before.
[$_{\text{VoiceP}}$ again [$_{\text{VoiceP}}$ Lili [$_{\text{vP}}$ cut that big pancake]]]

c. agentless presupposition reading: Someone (else) had cut the big pancake before.
[$_{\text{VoiceP}}$ Lili [$_{\text{vP}}$ again [$_{\text{vP}}$ cut that big pancake]]]

(adapted from Zhang, 2022, p. 94, ex. (22))

If such a structural analysis of the agentless presupposition reading is on the right track, children who access this reading must already have a basic understanding of the repetitive meaning of 'again', which aligns with the results of the Felicity Judgment Task.[18] To our knowledge, no prior study has investigated children's knowledge of this reading, making it a topic for future research.

5.4. On children's ability to project presuppositions from *wh*-questions

Another possible explanation for children's selection of the 'both' response in the Question-Answer Task involves the way they project presuppositions from *wh*-questions. Unlike adults, children may not restrict the domain of possible answers to those the speaker considers possibly true. Instead, they may access a universal projection interpretation, which leads to presupposition failure, leading them to ignore the trigger in the *wh*-questions.

Similar ideas have been proposed regarding the way children interpret presuppositions under quantifiers like "none". Zehr et al. (2016) examined how adults and children interpret sentences like "None of the bears won the race", which contains the presupposition trigger "win" (presupposing that the relevant characters *participated* in the race). Their findings indicate that while adults access both the universal projection interpretation (i.e., *every bear participated in the race*) and the existential projection interpretation (i.e., *at least one of the bears participated in the race*), children only access the universal projection interpretation. To reconcile theories of presupposition projection with the experimental data from both adults and children, the authors propose that for presupposition projection from the scope of negative quantificational expressions,

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selection of the 'again speaker' in control trials. While the narrative progression of the stories could in principle make the agentless presupposition reading available, the contrastive nature of the Felicity Judgment Task would likely have directed participants' attention to the single difference between the two alternative descriptions: the presence vs. absence of "you". Because both descriptions identified only one agent, the agentless presupposition reading would have been less salient. In any case, as reported in Section 5.1, only two (out of 40) children consistently selected the 'again speaker' on control trials.

[18] Note that the results of our two experiments cannot, on their own, determine which reading is basic and which is derived. The availability of the agentless presupposition reading, as well as its derivation, has been proposed in the theoretical literature on the basis of independent evidence. We thank an anonymous reviewer for encouraging us to make this explicit.

universal projection is obtained directly via projection, while existential projection builds upon it and requires additional mechanisms, such as domain restriction, which the child participants may not yet have mastered. Our experimental findings corroborate this proposal.

If our explanation is on the right track, we predict that children might also ignore other presupposition triggers (such as "also") in similar tasks, where universal presupposition projection in *wh*-questions leads to presupposition failure. However, this prediction does not appear to be borne out by the results reported in Kurokami et al. (2021, 2022) and Kurokami (2025), who used a similar Question-Answer Task. In their studies, participants first listened to stories in which Mickey, Minnie, and Donald each performed one or two actions. For example, Minnie ate only an apple, Donald only a banana, and Mickey both. At the end of each story, a puppet was asked who ate the most fruit. After recalling part of the story ("Donald didn't eat an apple, but Mickey and Minnie did."), the forgetful puppet asked a *wh*-question to solicit help from the participant, such as "Who (also) ate a banana?" The results showed that children were more likely to select the character who performed both actions in the "also" condition than in the condition without "also".[19] Crucially, the proportion of adult-like responses, i.e., child responses selecting the two-action character in the critical trials (41% in the Japanese experiment, 52%-70% in the English experiments) was higher than the proportion of adult-like performance in our Question-Answer Task, which yielded only 25% of child responses selecting the repeater in the critical "again" trials.

A closer examination reveals two key differences between Kurokami et al.'s experiments and ours. First, their *wh*-question was not a non-canonical information-seeking question. Rather, the forgetful puppet genuinely sought information from participants. Second, the *wh*-question was embedded within a larger discourse, where the puppet's partial recall of the story (Mickey and Minnie ate apples) served as a linguistic antecedent that satisfied the presupposition of "also" in the subsequent question 'Who also ate a banana?' This prior discourse might have restricted the domain of potential answers to the two characters who satisfied the presupposition of "also". Even if children interpreted the presupposition as projecting universally, there was no presupposition failure. As a result, Kurokami et al.'s experimental findings are not inconsistent with our predictions. Future research could more closely examine whether children differ from adults in how they project presuppositions from *wh*-questions.

6. Conclusion

In this study, we implemented two alternative tasks to the TVJT—a Question-Answer Task (Experiment 1) and a Felicity Judgment Task (Experiment 2)—which were both designed to tap into preschool-aged children's knowledge of the presupposition of Mandarin "you" ('again'). Both tasks avoided the problem of forcing participants to make binary judgments (e.g., 'true'/'false', 'yes'/'no') about sentences exhibiting presupposition failure. Importantly, the results hold without having to exclude children who might appear to ignore the trigger. In contrast to previous findings, the Question-Answer Task showed children responding as if they were ignoring the trigger. We discussed several factors which might have led to their non-adultlike behavior. A follow-up Felicity Judgment Task, which was designed to address some of the issues with the first experiment, revealed that children successfully understood the presuppositional trigger (albeit not at fully

[19] Notably, while Kurokami et al. (2021) reported relatively low proportions of child responses selecting the two-action character in their critical "also" condition (41% in the Japanese experiment, 52% in the English experiment), their follow-up study with methodological improvements (Kurokami et al., 2022) yielded a significantly higher accuracy rate of 70%.

adult-like levels). These results suggest that children's non-adultlike behavior in the Question-Answer Task should not be attributed to a lack of knowledge of "you".

Future research might apply a similar paradigm to evaluate children's knowledge of other presuppositional triggers similar to "you" ('again'). More generally, we hope the present study offers methodological insights for investigating presuppositions and helps pave the way for further work exploring ways of felicitously tapping into children's knowledge of presupposition triggers.

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Appendix

Here we present visualizations of children's selections of each response option across different age groups in Experiment 1 and Experiment 2.

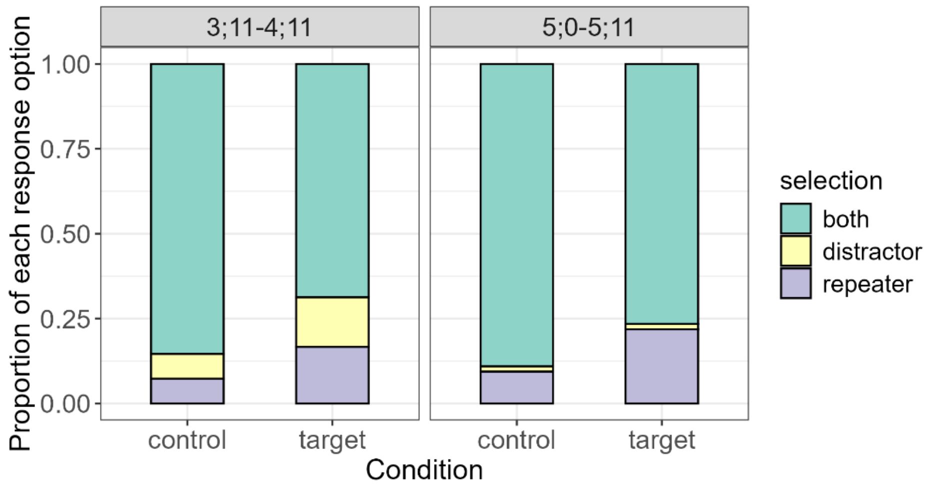

Figure 7: Proportion of children's selections of each response option across different age groups in Experiment 1.

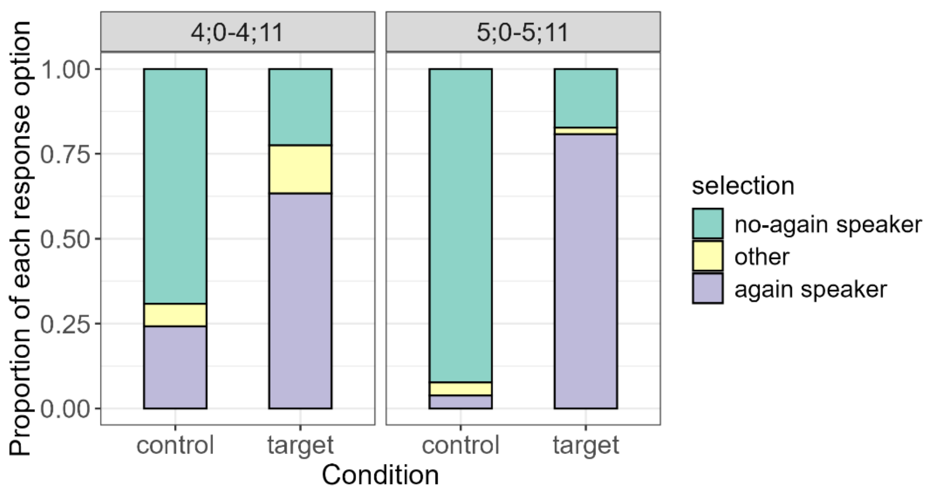

Figure 8: Proportion of children's selections of each response option across different age groups in Experiment 2.